

431 Lab 4 Instructions (REVISED 2026-09-17)

Fall 2026 - deadline in [Course Calendar](#)

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2026-09-17

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! Important

- This Lab contains 3 tasks for you to complete and was revised on 2026-09-17.
- The deadline for completing this Lab is posted in the [Course Calendar](#).

0.1 R Setup

```
library(tidyverse)
```

You will use additional packages in developing your response.

We have built no template for Lab 4. Instead, use the templates for Lab 1 and Lab 2 we have provided to help you get started.

0.2 Getting Help

You may discuss each Lab with Professor Love, the teaching assistants or your colleagues, but your answer must be prepared by **you working alone**. Don't be afraid to ask questions, using any of the methods described on [our Contact Us page](#).

0.3 Learning Objectives for this Lab

1. Ingest data from an Excel (.xlsx) file into R
2. Complete appropriate analyses of multiple two-way contingency tables.
3. Build appropriate summaries of a single predictor linear regression model.
4. Communicate effectively about your results in complete, clear English sentences.

1 Task 1 (10 points)

1.1 The Data

For Tasks 1 and 2, we use the data in the `lab4-task1.xlsx` (Excel) file, which is derived from the LicoriceGargle example at the [Cleveland Clinic Statistical Dataset Repository](#) which provides two files to explain the data.

- a [general description](#), and
- a [data description](#).

The data were originally published in Ruetzler K et al. (2013) [A randomized, double-blind comparison of licorice versus sugar-water gargle for prevention of postoperative sore throat and postextubation coughing](#) *Anesth Analg* 117: 614-21. <https://doi.org/10.1213/ane.0b013e318299a650>.

From the Abstract to the Ruetzler paper, we have the following details:

We enrolled 236 patients having elective thoracic surgery who required intubation with a double-lumen endotracheal tube. Patients were randomly assigned to gargle 5 minutes before induction of anesthesia for 1 minute with: (1) Extractum Liquiritiae Fluidum (licorice 0.5 g); or (2) Sirupus Simplex (sugar 5 g); each diluted in 30 mL water.

Our small sample of the original data includes only 235 rows and three variables:

Variable	Description
<code>subject</code>	Subject ID, added by Dr. Love
<code>intervention</code>	sugar or licorice (from <code>treat</code> in the CCF dataset)
<code>postop4</code>	Yes or No regarding sore throat pain at rest 4 hours after surgery (from <code>postOp4hour_throatpain</code>)

1.2 The Task

Compare the rates of post-operative sore throat pain among subjects who gargled with licorice as compared to those who gargled with sugar. In particular, create and share an appropriate statistical summary of the relationship, and use appropriate statistical methods to estimate the point estimate and 95% confidence intervals for an appropriate:

- relative risk,
- odds ratio, and
- risk difference.

In this case, do not use a Bayesian augmentation.

2 Task 2 (15 points)

Reflecting on what you've done in Task 1 and what you've read in Spiegelhalter's *The Art of Statistics*, please write an essay of 125-200 words which discusses the relationship between `postop4` and `intervention` that you observe in this sample.

In your essay, first, describe what you have learned about the association of licorice (vs. sugar) and sore throat pain 4 hours post-surgery from this work, avoiding jargon and also avoiding statements about statistical significance. Then, discuss how what we know about the study design tells us about what we might be able to conclude about whether gargling with licorice reduces pain at 4 hours post-surgery.

3 Task 3 (25 points)

3.1 The Data

The `lab4-task3.Rds` dataset (also known as the *lindner* data) provided on our [431-data page](#) comes from an observational study of 996 patients receiving an initial Percutaneous Coronary Intervention (PCI) at Ohio Heart Health, Christ Hospital, Cincinnati in 1997 and followed for at least 6 months by the staff of the Lindner Center.

The 698 patients thought to be more severely diseased were assigned to treatment with **abciximab** (an expensive, high-molecular-weight IIb/IIIa cascade blocker); while the remaining 298 patients received **usual care** with their initial PCI.

Additional information on the [lindner data set is available here](#).

These data relate to Kereiakes DJ, Obenchain RL, Barber BL, et al. Abciximab provides cost effective survival advantage in high volume interventional practice. *Am Heart J* 2000; 140: 603-610.

3.2 The Task, in Five Parts

Ingest the `lab4_task3.Rds` data into R, and use them to develop an appropriate linear regression model to predict `cardbill` (which you can interpret as cardiac-related costs, in US dollars) on the basis of `ejecfrac` (left ventricular ejection fraction, measured as a percentage.)

This task has five parts.

- a. (5) Produce a scatterplot of the relationship, with the natural logarithm of `cardbill` on the vertical (Y) axis, and `ejecfrac` on the horizontal (X) axis, including a blue loess smooth and a red linear model fit. In a sentence or two, comment on whether the straight line model matches the data well.

Hints for part a

You can use either `geom_point()` or `geom_jitter()` here.

- b. (5) Build an ordinary least squares model predicting the natural logarithm of `cardbill` using `ejecfrac`, and specify and interpret the fitted equation in a sentence or two in context of the problem.

💡 Hint for parts b and c

Exponentiate the coefficients of your model. Also, ask for three decimal places by including `digits = 3` within your call to `model_parameters()`.

- c. (5) Provide the 90% confidence interval for the slope of your regression line, and use a sentence or two to describe it carefully and correctly.
- d. (4) Obtain the R^2 and σ values for the model you fit in part b, and interpret each of them in context of this study in a complete sentence or two.

💡 Hint for part d

Remember the outcome that your model is predicting.

- e. (6) Use `check_model()` to examine the posterior predictive check and the Normality of residuals for your model and identify any potential problems with those two plots in complete sentences.

💡 Hints for part e

- Most of the points for this question require you to assess whether the model you fit in part b shows signs of trouble in either (or both) of these plots.
- *If* the plots you produce are hard to read in your HTML, then use `# fig-height: 8` at the start of your code chunk creating the plots to make them taller.
- One additional problem you *may* observe is that the x-axis of the posterior predictive check may in fact be mislabeled. If your plot is mislabeled in this regard, specify what the correct label should be as part of your response.

4 Section 4 of your Lab Report: AI Usage

All students should include an AI Usage section in each assignment for this class. See the instructions from Lab 1 for more details.

5 Section 5 of your Lab Report: Session Information

Include the session information as a final section in this Lab. I've done so at the bottom of this document.

6 Additional Notes and Instructions

6.1 Submitting this Lab

Submit this Lab via [Canvas](#), using the Lab 4 assignment. Be sure to submit both:

1. Your Quarto file (.qmd), and
2. The HTML file you obtain by knitting the Quarto file (.html)

Be sure that your Quarto (and thus HTML) files include the **AI Usage** and **Session information** sections at the end of the document, and that these section headings appear in the Table of Contents.

Be sure that your Quarto (and thus HTML) files include the **AI Help** and **Session information** sections at the end of the document, and that these section headings appear in the Table of Contents.

6.2 Grading this Lab

This Lab will be graded by the TAs and then reviewed by Dr. Love. Your grades will be available one week after the Lab deadline.

The maximum score on this Lab is 50 points.

As each Lab passes its deadline (as listed in the [Course Calendar](#)), we will:

- post the answer sketch (48 hours after the deadline) and draft grading rubric to our Shared Google Drive, and then
- post grades and any revisions to the grading rubric or answer sketch one week after the deadline to a location we will provide to you.

6.3 Emergencies and Late Policy

We do not grant extensions on Lab deadlines.

- To receive full credit on a Lab, it must be received on Canvas no later than 59 minutes after the posted deadline. (This allows for small issues with uploading to Canvas to occur without penalty.)
 - Labs that are turned in 1-48 hours after the deadline will lose 10 points for late work.
- No extensions to Lab deadlines will be made this semester. Labs turned in more than 48 hours after the deadline will receive no credit, since by then the Lab Sketch will be posted.
- For details on how Lab grades affect your course grade, see the Syllabus.
- If you have an emergency that will keep you from submitting the Lab by even the late deadline of Friday at noon, please let Dr. Love know that (as soon as possible) via email and he will consider excusing you from the Lab.

6.4 Lab Regrade Requests

If, after your Lab is graded, you want Dr. Love to review the grading or correct a grading error, please follow the Lab Regrade Request policy [posted on our Labs page](#).

7 Session Information

```
xfun::session_info()
```

```
R version 4.6.1 (2026-06-24 ucrt)  
Platform: x86_64-w64-mingw32/x64  
Running under: Windows 11 x64 (build 26200)
```

Locale:

```
LC_COLLATE=English_United States.utf8  
LC_CTYPE=English_United States.utf8  
LC_MONETARY=English_United States.utf8  
LC_NUMERIC=C  
LC_TIME=English_United States.utf8
```

Package version:

askpass_1.2.1	backports_1.5.1	base64enc_0.1.6
bit_4.6.0	bit64_4.8.6	blob_1.3.0
broom_1.0.13	bslib_0.12.0	cachem_1.1.0
callr_3.8.0	cellranger_1.1.0	cli_3.6.6
clipr_0.8.1	compiler_4.6.1	conflicted_1.2.0
cpp11_0.5.5	crayon_1.5.3	curl_8.0.0
data.table_1.18.6.1	DBI_1.3.0	dbplyr_2.6.0
digest_0.6.39	dplyr_1.2.1	dtplyr_1.3.3
evaluate_1.0.5	farver_2.1.2	fastmap_1.2.0
fontawesome_0.5.3	forcats_1.0.1	fs_2.1.0
gargle_1.6.1	generics_0.1.4	ggplot2_4.0.3
glue_1.8.1	googledrive_2.1.2	googlesheets4_1.1.2
graphics_4.6.1	grDevices_4.6.1	grid_4.6.1
gtable_0.3.6	haven_2.5.5	highr_0.12
hms_1.1.4	htmltools_0.5.9	httr_1.4.9
ids_1.0.1	isoband_0.3.0	jquerylib_0.1.4
jsonlite_2.0.0	knitr_1.52	labeling_0.4.3
lifecycle_1.0.5	lubridate_1.9.5	magrittr_2.0.5
memoise_2.0.1	methods_4.6.1	mime_0.13
modelr_0.1.11	openssl_2.4.2	otel_0.2.0
pillar_1.11.1	pkgconfig_2.0.3	prettyunits_1.2.0
processx_3.9.0	progress_1.2.3	ps_1.9.3
purrr_1.2.2	R6_2.6.1	ragg_1.5.2
rappdirs_0.3.4	RColorBrewer_1.1-3	readr_2.2.0
readxl_1.5.0	rematch_2.0.0	rematch2_2.1.2
reprex_2.1.1	rlang_1.3.0	rmarkdown_2.32
rstudioapi_0.19.0	rvest_1.0.5	S7_0.2.2

sass_0.4.10	scales_1.4.0	selectr_0.6.0
stats_4.6.1	stringi_1.8.9	stringr_1.6.0
sys_3.4.3	systemfonts_1.3.2	textshaping_1.0.5
tibble_3.3.1	tidyr_1.3.2	tidyselect_1.2.1
tidyverse_2.0.0	timechange_0.4.0	tinytex_0.60
tools_4.6.1	tzdb_0.5.0	utf8_1.2.6
utils_4.6.1	uuid_1.2.2	vctrs_0.7.3
viridisLite_0.4.3	vroom_1.7.1	withr_3.0.3
xfun_0.60	xml2_1.6.0	yaml_2.3.12